

A PROJECT REPORT ON

JobInGen Trend-Grounded Marketing Agent

An Autonomous Multi-Agent System for Trend-Driven Social Media Content Generation

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Abstract

Social media content for career-guidance brands is usually written in isolation from what the audience is actually discussing, which makes it generic and easy to ignore. This project, JobInGen Trend-Grounded Marketing Agent, addresses this gap by building an automated system that first studies live online conversation — on Reddit, Google Search trends, and career-related news — before writing a single line of marketing copy. The system identifies topics that students and early job seekers are genuinely concerned about, evaluates how relevant and fast-rising each topic is, and then drafts platform-specific posts for LinkedIn and Instagram that connect the trending topic to JobInGen's product capabilities such as resume tailoring, mock interviews, and career guidance.

The application is implemented in Python and structured as a multi-stage agentic pipeline using LangGraph for orchestration and Pydantic for enforcing strict data formats between stages. Google's Gemini API supplies the language understanding and generation capability used to analyse trends, decide on a content angle, and write the final post copy. Live signal is gathered through pytrends (Google Trends), the News API, and PRAW (the Python Reddit API Wrapper). Each generated post passes through an automated review stage that checks platform fit, brand safety, and predicted engagement before being marked ready for human approval. The result is a working prototype that demonstrates how a research-first, multi-agent design can produce content that is timely, audience-aware, and platform-appropriate, rather than generic AI-written text.

1. Introduction

Most brands that try to automate their social media struggle with one consistent problem: the content sounds disconnected from reality. A piece of writing produced purely from a generic instruction such as “write a post about job interviews” tends to read as bland and forgettable because it is not grounded in anything the audience is currently experiencing. Human marketers avoid this trap instinctively — they read the room before they write. They notice what people are complaining about online, what topic is suddenly being searched more, and what is in the news that week, and they shape their content around that real conversation.

This project attempts to encode that instinct into software. JobInGen is positioned as an AI-driven career platform for students and early-career job seekers, offering tools such as an ATS-friendly resume builder, an adaptive mock-interview simulator, and a career-guidance copilot. The marketing agent built in this project is designed to support JobInGen's social media presence by automatically discovering what its audience is anxious or excited about right now, and converting that insight into ready-to-review posts for LinkedIn and Instagram.

Rather than treating content generation as a single prompt-to-output task, the system is designed as a pipeline of cooperating components (referred to as agents), each responsible for a narrow part of the overall reasoning: gathering raw signal, identifying themes, judging audience relevance, choosing a differentiated angle, writing platform-specific copy, and finally reviewing the draft before it reaches a human reviewer. This division of labour mirrors how a small marketing team would actually operate, and it is the central design idea explored in this project.

2. Problem Statement

Manually researching trending topics, drafting platform-appropriate copy, and tailoring messaging for two very different platforms (LinkedIn and Instagram) is time-consuming and difficult to do consistently, especially for a small team. Existing AI writing tools generate text quickly but typically work from a topic given by a human and have no built-in mechanism to confirm that the topic is actually relevant or currently trending. This leads to two recurring failures in practice:

- Content that feels generic because it is not grounded in any verifiable, real conversation happening on social platforms or in the news.
- Content that ignores the fact that LinkedIn and Instagram audiences expect very different tone, structure, and length, so a single piece of writing reused across both platforms under-performs on at least one of them.

The problem this project sets out to solve is therefore: how can a software system autonomously discover genuinely trending, audience-relevant topics from real online sources, and translate them into distinct, platform-appropriate marketing posts, while keeping a human reviewer in control of what is finally published?

3. Objectives

- To design and implement a multi-agent pipeline using LangGraph that automates the journey from raw trend signal to a finished, reviewable social media post.
- To collect real-time signal from Reddit (via PRAW), Google Trends (via pytrends), and career/hiring news (via the News API).
- To use the Gemini API for natural-language reasoning tasks: clustering raw posts into themes, judging audience relevance, selecting a content angle, and writing the final copy.
- To enforce strict, type-checked data exchange between pipeline stages using Pydantic models, so that malformed output is caught early instead of silently breaking the pipeline.
- To generate genuinely different content for LinkedIn and Instagram from the same underlying trending topic, reflecting each platform's distinct reader behaviour.
- To build an automated quality-check stage that scores each draft on predicted engagement before it is shown to a human reviewer for final approval.

4. Proposed System

The proposed system is a single Python application that runs as a directed pipeline of processing stages, implemented using the LangGraph state-machine library. Each run begins with a trigger — either a manual request or a background check that finds a topic strong enough to act on. The system then moves through four broad stages: gathering signal from external sources, reasoning about what that signal means for the target audience, generating platform-specific written content, and finally reviewing that content automatically before it is handed to a human for approval.

Unlike a simple “prompt-and-generate” tool, the proposed system treats research as a first-class step that happens before any writing occurs. It also keeps a human firmly in the loop: nothing is published automatically. Every accepted draft is accompanied by a short analysis explaining why the topic was chosen and how the post is expected to perform, so the reviewer can make an informed decision quickly.

5. Key Features

- Live signal collection from Reddit discussions, Google Trends search data, and hiring-related news articles.
- Automatic clustering of scattered posts and headlines into a small number of coherent, named themes.
- A numeric trend score combining audience relevance, how fast a topic is rising, and how well it fits the JobInGen brand, used to decide whether a topic is worth posting about.
- A built-in “no post today” outcome — if nothing scores high enough, the system deliberately stays silent instead of forcing out a weak post.
- Separate content briefs and writing styles for LinkedIn and Instagram, reflecting each platform’s different reading behaviour.
- Structured, machine-checked output (hook lines, body copy, caption, hashtags, call-to-action, and image alt-text) validated through Pydantic models.
- An automated review stage that screens drafts for platform fit, brand-safety violations (such as exaggerated promises), and predicted engagement.
- A mandatory human-approval checkpoint before any content is considered final.

6. System Architecture

The system is organised into four logical layers. Each layer is a group of components that share a common purpose, and information flows downward from one layer to the next. If a later layer detects a problem with the content (for example, a weak opening line), the system routes the task back to the specific earlier layer responsible for that aspect, rather than restarting the whole process from scratch.

6.1 Layer 1 — Signal Collection

This layer is responsible for gathering raw, real information from the outside world. It pulls recent posts from selected subreddits using PRAW, checks rising search terms using pytrends, and retrieves recent hiring-related articles using the News API. The raw items collected here are then grouped into a handful of candidate themes, each backed by real evidence (post snippets, links, and a measure of how active the topic currently is).

6.2 Layer 2 — Strategy

Once a theme is selected, this layer figures out what the content should actually say and to whom. It identifies the underlying emotion the audience is likely to feel about the topic (for example, anxiety about interview performance), chooses a content angle that is specific to

JobInGen rather than generic career advice, and decides how that angle should be adapted differently for LinkedIn versus Instagram.

6.3 Layer 3 — Content Creation

This layer turns the strategic brief into actual written and visual content: hook lines, the main body or slide text, a caption, hashtags, and a call-to-action. A simple visual-selection step also decides what kind of accompanying image format (a clean text-based graphic, an illustrative image, or a data chart) best fits the post.

6.4 Layer 4 — Review and Analysis

Before anything reaches a human reviewer, the draft is automatically scored against platform expectations, brand-safety rules, and a five-factor predicted-performance model covering hook strength, trend alignment, emotional relevance, value density, and likelihood of being saved or shared. If a draft fails a check, the relevant earlier layer is asked to revise only the failing part. Once a draft passes, it is packaged with a short analysis report and sent for human approval.

Diagram placeholder: insert a four-box vertical flow diagram here titled “Four-Layer Architecture of the JobInGen Marketing Agent”, showing Layer 1 (Signal) at the top flowing into Layer 2 (Strategy), then Layer 3 (Creation), then Layer 4 (Review/Analysis), with a feedback arrow looping back from Layer 4 to the relevant earlier layer.

Layer	Purpose	Main Activity
Layer 1 – Signal	Gather real-world data	Pull Reddit posts, Google Trends data, and news articles; group into themes
Layer 2 – Strategy	Decide what to say	Identify audience emotion, choose a differentiated angle, plan per-platform approach
Layer 3 – Creation	Produce the content	Write hooks, captions, hashtags, CTA; select visual format
Layer 4 – Review	Check quality before publishing	Score platform fit, brand safety, and predicted engagement; route failures back

7. Workflow of the Application

A single run of the system proceeds through the following steps, implemented as nodes in a LangGraph state graph:

- A run is triggered manually or automatically when background monitoring detects a sufficiently strong trend.
- Recent posts, search trends, and news items are pulled and tagged with their source and a recency/score value.
- Raw items are clustered into a small number of named themes, each supported by representative (paraphrased) evidence.

- Each theme is scored on audience relevance, how quickly it is rising, and brand fit; the strongest theme is selected, or the run ends with “no strong theme”.
- The underlying audience emotion connected to the chosen theme is identified.
- A differentiated content angle is chosen that ties back to a real JobInGen feature wherever possible.
- Separate creative briefs are produced for LinkedIn and Instagram.
- The copy (hooks, body, caption, hashtags, CTA, alt-text) is generated for each platform.
- A visual format is chosen and an image is produced or composed for each post.
- Automated reviewers score the draft; failing drafts are sent back to the responsible stage for revision (within a limited number of retries).
- Approved packs are presented to a human reviewer along with a short analysis report explaining the reasoning and predicted performance.

Text-based workflow diagram:

```

START
|
v
[ Collect Signal: Reddit / Trends / News ]
|
v
[ Cluster into Themes + Score Themes ]
|
v
[ No strong theme? ] --YES--> [ END: no post this run ]
| NO
v
[ Identify Audience Emotion ]
|
v
[ Select Differentiated Angle ]
|
v
[ Build LinkedIn Brief ]    [ Build Instagram Brief ]
|                          |
v                          v
[ Write Copy ]              [ Write Copy ]
|                          |
v                          v
[ Select / Build Visual ]    [ Select / Build Visual ]
|                          |
+-----+-----+
|
v
[ Automated Review: Platform Fit, Brand Safety, Engagement Score ]
|
fail | pass
<----- back to relevant stage
v
[ Human Approval ] --> [ END: post pack delivered ]

```

8. Technology Stack

The project is built entirely in Python, using the following libraries and services:

Technology	Role in the Project
Python	Core programming language used to implement the entire application.
LangGraph	Orchestrates the multi-stage pipeline as a state graph, including conditional routing and retries.
Pydantic	Defines and validates the structured data passed between pipeline stages, preventing malformed output.
Google Gemini API	Provides the language-model reasoning used for theme synthesis, angle selection, copywriting, and review scoring.
pytrends	Unofficial Python client used to read Google Trends data and detect rising search interest.
News API	Retrieves recent hiring and career-related news articles to add macro context to trending themes.
PRAW (Python Reddit API Wrapper)	Accesses Reddit's API to collect posts and discussions from relevant subreddits.
python-dotenv	Loads API keys and configuration values from a local .env file, keeping secrets out of source code.
Python Virtual Environment (venv)	Isolates project dependencies from the system-wide Python installation.

9. Module Description

The codebase is organised into modules that mirror the four-layer architecture described earlier. The table below summarises the main modules and their responsibilities.

Module	Responsibility
Signal Adapters (reddit.py, trends.py, news.py)	Connect to PRAW, pytrends, and the News API respectively, and return a common, normalised data format.
Theme Synthesizer	Groups raw signal items into candidate themes with supporting evidence.
Trend Scorer	Calculates a numeric score for each theme and selects the strongest candidate, or returns “no strong theme.”
Audience Modeler	Determines the core emotional tension the audience feels about the chosen theme.
Angle Selector	Chooses a differentiated content angle tied to a JobInGen product feature where possible.
Platform Strategist	Builds separate creative briefs for LinkedIn and Instagram from the same angle.
Copywriter	Generates the structured post copy: hooks, body, caption, hashtags, CTA, and alt-text.
Visual Director	Selects and triggers the appropriate visual format for the post.

Module	Responsibility
Review Module (Critics)	Automatically checks platform fit, brand safety, and predicted engagement; routes failures back.
State Models (state.py)	Pydantic models defining the shared data structure passed between all pipeline stages.
Configuration (config.yaml, .env)	Stores scoring weights, thresholds, and API credentials used across the system.

10. Advantages

- Content is grounded in verifiable, real-time conversation rather than invented from a generic prompt, making it more relevant and timely.
- Separate treatment of LinkedIn and Instagram ensures each post respects the reading habits of its specific audience instead of reusing one piece of text everywhere.
- The structured, typed data contracts (via Pydantic) catch malformed or incomplete content early, preventing broken posts from reaching the reviewer.
- The automated review stage filters out weak hooks, off-brand claims, and exaggerated promises before a human ever sees the draft, saving review time.
- The system can choose not to post when nothing is genuinely trending, which avoids diluting the brand's feed with low-quality, forced content.
- The modular, layer-based design makes it straightforward to add new signal sources or new platforms in the future without rewriting the whole pipeline.

11. Future Scope

- Adding a learning loop that compares predicted engagement scores against actual post performance metrics, to gradually improve the scoring model.
- Extending the system to additional platforms such as X (Twitter) or YouTube Shorts/Community posts.
- Building a lightweight dashboard for reviewers to approve, edit, or reject posts directly instead of working from raw output files.
- Integrating JobInGen's own internal usage data (e.g., anonymised resume or interview statistics) as an additional, first-party signal source.
- Adding direct publishing integrations for LinkedIn and Instagram so approved posts can be scheduled automatically after human sign-off.
- Replacing the simple polling-based trigger with a more responsive, event-driven mechanism for detecting sudden spikes in conversation.

12. Conclusion

This project demonstrates that a research-first, multi-agent approach can meaningfully improve the quality and relevance of automatically generated social media content. By collecting real signal from Reddit, Google Trends, and news sources before any writing takes place, and by enforcing strict, validated data exchange between pipeline stages using Pydantic and LangGraph, the system avoids the generic, disconnected tone common in simple AI-generated marketing text. Treating LinkedIn and Instagram as genuinely different platforms, rather than resizing one post for both, produces content that is more likely to match audience expectations on each. Keeping a human reviewer at the final approval step ensures the system remains a productivity aid rather than an uncontrolled publishing pipeline. Overall, the project achieves its objective of building a working prototype that turns live online conversation into grounded, platform-aware, and reviewable marketing content for JobInGen.

13. References

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